

Markscheme

May 2018

Chemistry

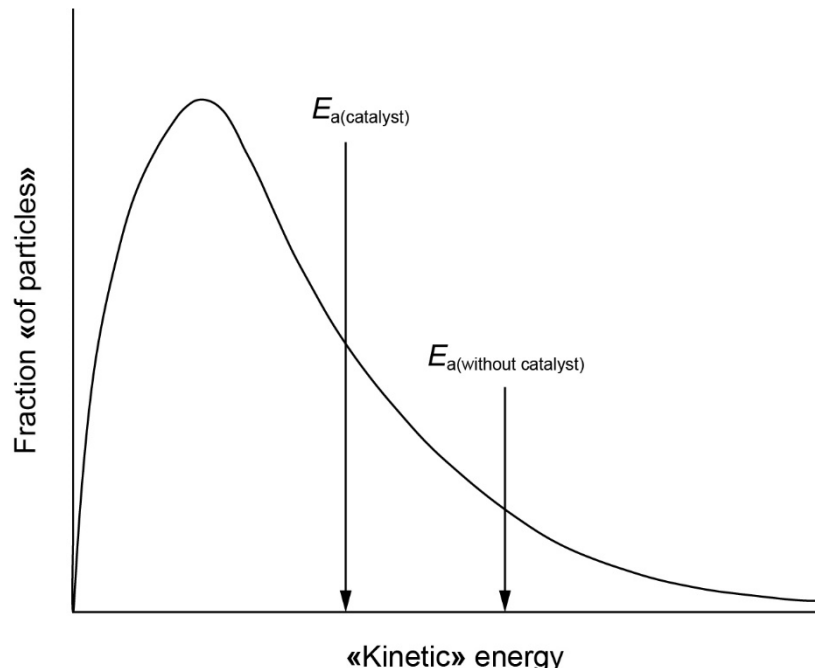
Higher level

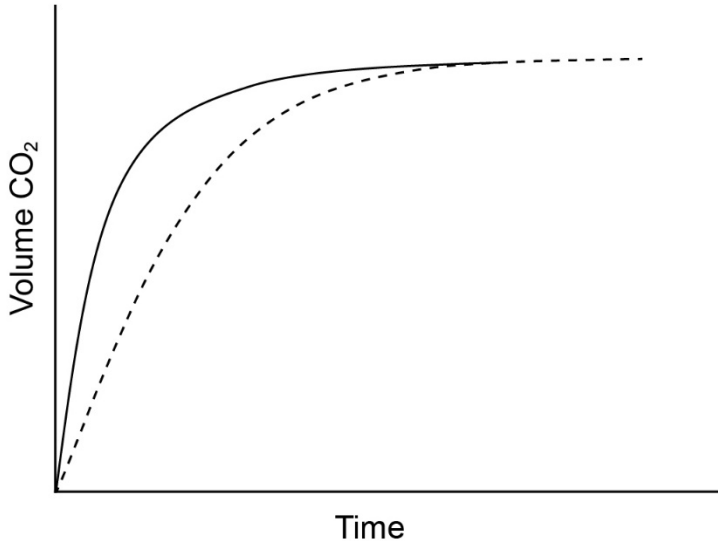
Paper 2

22 pages

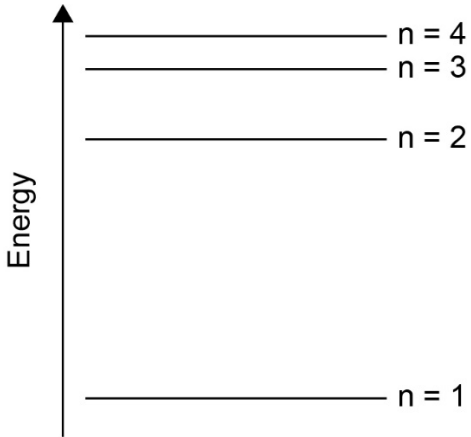
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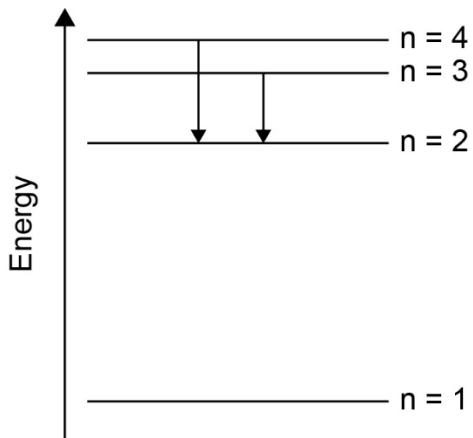
Question			Answers	Notes	Total
1.	a		$n(\text{H}_2\text{SO}_4) \llcorner = 0.0500 \text{ dm}^3 \times 0.100 \text{ mol dm}^{-3} \llcorner = 0.00500 \times 10^{-3} \llcorner \text{mol} \llcorner \checkmark$		1
1.	b		$\text{H}_2\text{SO}_4(\text{aq}) + \text{Al}(\text{OH})_3(\text{s}) \rightarrow \text{MgSO}_4(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) \checkmark$	Accept an ionic equation.	1
1.	c		$\llcorner n(\text{H}_2\text{SO}_4) = \frac{1}{2} \times n(\text{NaOH}) = \frac{1}{2} (0.02080 \text{ dm}^3 \times 0.1133 \text{ mol dm}^{-3}) \llcorner$ $\llcorner \times 10^{-3} \llcorner \text{mol} \llcorner \checkmark$		1
1.	d		$n(\text{H}_2\text{SO}_4) \text{ reacted } \llcorner = 0.00500 - 0.001178 \llcorner = 0.00382 \times 10^{-3} \llcorner \text{mol} \llcorner \checkmark$		1
1.	e		$m(\text{Al}(\text{OH})_3) \llcorner n(\text{H}_2\text{SO}_4) = \llcorner = 0.00382/3.82 \times 10^{-3} \llcorner \text{mol} \llcorner \checkmark$ $m(\text{Al}(\text{OH})_3) \llcorner = 0.00382 \text{ mol} \times 78.00 \text{ g mol}^{-1} \llcorner = 0.0948 \text{ g} \llcorner \checkmark$	Award [2] for correct final answer.	2
1.	f		$\% \text{ Al}(\text{OH})_3 \llcorner = \frac{0.0948}{1.621} \times 100 \llcorner = 5.85 \llcorner \checkmark$	Answer must show three significant figures.	1
1.	g		to reduce random errors OR to increase precision $\checkmark$	Accept "to ensure reliability".	1

Question			Answers	Notes	Total
2.	a		 both axes correctly labelled ✓ correct shape of curve starting at origin ✓ $E_{a(catalyst)} < E_{a(without catalyst)}$ on x-axis ✓	M1: Accept “speed” for x-axis label. Accept “number of particles”, “N”, “frequency” or “probability «density»” for y-axis label. Do not accept “potential energy” for x-axis label. M2: Do not accept a curve that touches the x-axis at high energy. Do not award M2 if two curves are drawn. M3: Ignore any shading under the curve.	3

Question			Answers	Notes	Total
2.	b	i	 curve starting from origin with steeper gradient AND reaching same maximum volume ✓		1
2.	b	ii	rate decreases OR slower reaction ✓ « OR lower [H⁺] ✓ » partially dissociated/ionized «in solution/water» propanoic acid	Accept “weak acid” or “higher pH”.	2

Question			Answers	Notes	Total
2.	c		«pH» converts «wide range of $[H^+]$» into simple «log» scale/numbers OR «pH» avoids need for exponential/scientific notation OR «pH» converts small numbers into values «typically» between 0/1 and 14 OR «pH» allows easy comparison of values of $[H^+]$ ✓	Accept “uses values between 0/1 and 14”. Do not accept “easier to use”. Do not accept “easier for calculations”.	1
2.	d	i	A: CH_3COOH/ethanoic/acetic acid AND CH_3COO^- /acetate ions ✓ B: CH_3COO^-/ethanoate/acetate ions ✓	Penalize “sodium ethanoate/acetate” instead of “ethanoate/acetate ions” only once.	2
2.	d	ii	$K_a = 1.74 \times 10^{-5} = \frac{[H^+]^2}{0.10}$ OR $[H^+] = 1.32 \times 10^{-3}$ «mol dm⁻³» ✓ «pH =» 2.94 ✓	Accept [2] for correct final answer.	2
2.	d	iii	«forms weak acid and strong base, thus basic» $CH_3COO^-(aq) + H_2O(l) \rightleftharpoons CH_3COOH(aq) + OH^-(aq)$ ✓	Accept → for ⇌.	1
2.	d	iv	less than 7 ✓		1

Question			Answers	Notes	Total
2.	e	i	$2\text{NO}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{HNO}_2(\text{aq}) + \text{HNO}_3(\text{aq}) \checkmark$		1
2.	e	ii	$2\text{HNO}_2(\text{aq}) + \text{CaCO}_3(\text{s}) \rightarrow \text{Ca}(\text{NO}_2)_2(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$ OR $2\text{HNO}_3(\text{aq}) + \text{CaCO}_3(\text{s}) \rightarrow \text{Ca}(\text{NO}_3)_2(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) \checkmark$		1
3.	a	i	 4 levels showing convergence at higher energy $\checkmark$		1

Question			Answers	Notes	Total
3.	a	ii	 arrows (pointing down) from $n = 3$ to $n = 2$ AND $n = 4$ to $n = 2$ ✓		1

(continued...)

(Question 3a continued)

Question			Answers	Notes	Total						
3.	a	iii	$IE \ll \Delta E = h \nu = 6.63 \times 10^{-34} \text{ J s} \times 3.28 \times 10^{15} \text{ s}^{-1} = 2.17 \times 10^{-18} \text{ «J» } \checkmark$		1						
3.	a	iv	$\ll \lambda = \frac{c}{\nu} = \frac{3.00 \times 10^8 \text{ ms}^{-1}}{3.28 \times 10^{15} \text{ s}^{-1}} = 9.15 \times 10^{-8} \text{ «m» } \checkmark$		1						
3.	b	i	same number of shells/«outer» energy level/shielding AND nuclear charge/number of protons/ Z_{eff} increases «causing a stronger pull on the outer electrons» $\checkmark$		1						
3.	b	ii	K^+ 19 protons AND Cl^- 17 protons OR K^+ has «two» more protons $\checkmark$ same number of electrons/isoelectronic «thus pulled closer together» $\checkmark$		2						
3.	c	i	<table><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>	1	1	1	1	1	1		1
1	1	1	1	1	1						
3.	c	ii	<i>Anode (positive electrode):</i> $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^- \checkmark$ <i>Cathode (negative electrode):</i> $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s) \checkmark$	<i>Accept $Cu(s) - 2e^- \rightarrow Cu^{2+}(aq)$.</i> <i>Accept $\rightleftharpoons$ for $\rightarrow$.</i> <i>Award [1 max] if the equations are at the wrong electrodes.</i>	2						

(continued...)

(Question 3c continued)

Question			Answers	Notes	Total
3.	c	iii	«external» circuit/wire AND from positive/anode to negative/cathode electrode ✓	Accept “through power supply/battery” instead of “circuit”.	1
3.	c	iv	no change «in colour» ✓	Do not accept “solution around cathode will become paler and solution around the anode will become darker”.	1
3.	c	v	oxygen/O ₂ ✓	Accept “carbon dioxide/CO ₂ ”.	1
3.	d		<i>Transition metals:</i> «contain» d and s orbitals «which are close in energy» OR «successive» ionization energies increase gradually ✓ <i>Alkali metals:</i> second electron removed from «much» lower energy level OR removal of second electron requires large increase in ionization energy ✓		2

Question			Answers	Notes	Total
4.	a		$\text{BrO}_3^- (\text{aq}) + 6\text{H}^+ (\text{aq}) + 6\text{I}^- (\text{aq}) \rightleftharpoons \text{Br}^- (\text{aq}) + 3\text{I}_2 (\text{s}) + 3\text{H}_2\text{O} (\text{l}) \checkmark$	Accept $\rightarrow$ for $\rightleftharpoons$.	1
4.	b		$n = 6 \checkmark$ $\llcorner \Delta G^\ominus = -nFE^\ominus \llcorner$ $\llcorner E^\ominus = -\frac{\Delta G^\ominus}{nF} = \frac{412 \times 10^3 \text{ J mol}^{-1}}{6 \times 9.65 \times 10^4 \text{ C mol}^{-1}} = \llcorner 0.711 \text{ V} \llcorner \checkmark$		2
4.	c		$\llcorner E^\ominus = E^\ominus (\text{BrO}_3^-/\text{Br}^-) - E^\ominus (\text{I}_2/\text{I}^-) \llcorner$ $\llcorner E^\ominus (\text{BrO}_3^-/\text{Br}^-) = E^\ominus + E^\ominus (\text{I}_2/\text{I}^-) = 0.888 + 0.54 = \llcorner 1.428 \text{ V} \llcorner \checkmark$		1

Question			Answers	Notes	Total
5.	a		bonds broken: $4(\text{C-H}) + 2(\text{H-O}) / 4(414) + 2(463) / 2582 \text{ «kJ» } \checkmark$ bonds made: $3(\text{H-H}) + \text{C}\equiv\text{O} / 3(436) + 1077 / 2385 \text{ «kJ» } \checkmark$ $\Delta H \llcorner = \sum \text{BE}_{(\text{bonds broken})} - \sum \text{BE}_{(\text{bonds made})} = 2582 - 2385 \llcorner = \llcorner + \llcorner 197 \text{ «kJ» } \checkmark$	Award [3] for correct final answer. Award [2 max] for -197 «kJ».	3
5.	b	i	$\Delta H_f^\ominus$ for any element = 0 «by definition» OR no energy required to form an element «in its stable form» from itself $\checkmark$		1
5.	b	ii	$\Delta H^\ominus \llcorner = \sum \Delta H_f^\ominus (\text{products}) - \sum \Delta H_f^\ominus (\text{reactants}) = -111 + 0 - [-74.0 + (-242)] \llcorner$ $= \llcorner + \llcorner 205 \text{ «kJ» } \checkmark$		1
5.	b	iii	«bond enthalpies» averaged values «over similar compounds» OR «bond enthalpies» are not specific to these compounds $\checkmark$		1
5.	c		$\Delta S^\ominus = \sum S^\ominus_{\text{products}} - \sum S^\ominus_{\text{reactants}} = 198 + 3 \times 131 - (186 + 189) \llcorner = \llcorner + \llcorner 216 \text{ «J K}^{-1}\llcorner \checkmark$		1
5.	d		$\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus = 205 \text{ kJ} - 298 \text{ K} \times \frac{216}{1000} \text{ kJ K}^{-1} \llcorner = \llcorner + \llcorner 141 \text{ «kJ» } \checkmark$		1

Question			Answers	Notes	Total
5.	e		$\Delta H^\ominus = T \Delta S^\ominus$ $T = \frac{\Delta H^\ominus}{\Delta S^\ominus} = \frac{205000 \text{ J}}{216 \text{ J K}^{-1}}$ $T = 949 \text{ K} \checkmark$	Do not award a mark for negative value of T.	1

Question			Answers	Notes	Total
6.	a		Q: non-equilibrium concentrations AND K_c : equilibrium concentrations OR Q: «measured» at any time AND K_c : «measured» at equilibrium ✓		1
6.	b		$Q = \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2 [\text{O}_2]} = \frac{1.00^2}{1.00^2 \times 2.00} = 0.500 \quad \checkmark$ reverse reaction favoured/reaction proceeds to the left AND $Q > K_c/0.500 > 0.500 \quad \checkmark$	Do not award M2 without M1.	2
6.	c	i	$[\text{N}_2\text{O}_2]$ decreases AND exothermic «thus reverse reaction favoured» ✓	Accept “product” for $[\text{N}_2\text{O}_2]$. Do not accept just “reverse reaction favoured/shift to left” for “ $[\text{N}_2\text{O}_2]$ decreases”.	1

(continued...)

(Question 6c continued)

Question			Answers	Notes	Total
6.	c	ii	ALTERNATIVE 1: «from equilibrium, step 1» $K_c = \frac{[\text{N}_2\text{O}_2]}{[\text{NO}]^2}$ OR $[\text{N}_2\text{O}_2] = K_c [\text{NO}]^2 \checkmark$ «from step 2, rate = $k_1 [\text{N}_2\text{O}_2][\text{O}_2] = k_2 K [\text{NO}]^2 [\text{O}_2]$» rate = $k [\text{NO}]^2 [\text{O}_2] \checkmark$ ALTERNATIVE 2: «from step 2» rate = $k_2 [\text{N}_2\text{O}_2] [\text{O}_2] \checkmark$ «from step 1, rate₍₁₎ = $k_1 [\text{NO}]^2 = k_{-1} [\text{N}_2\text{O}_2]$, $[\text{N}_2\text{O}_2] = \frac{k_1}{k_{-1}} [\text{NO}]^2$» «rate = $\frac{k_1}{k_{-1}} k_2 [\text{NO}]^2 [\text{O}_2]$» rate = $k [\text{NO}]^2 [\text{O}_2] \checkmark$	Award [2] for correct rate expression.	2
6.	d		«$\ln \frac{k_1}{k_2} = \frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$» $T_2 = \text{«}273 + 35 = \text{»}$ $T_1 = \text{«}273 + 25 = \text{»}$ $293 \text{ K} \checkmark$ $E_a = 75.8 \text{ «kJ mol}^{-1}\text{»} \checkmark$	Award [2] for correct final answer.	2

8

303 K AND T

Question			Answers	Notes	Total
7.	a	i	polar bonds «between H and group 16 element» OR difference in electronegativities «between H and group 16 element» ✓ uneven distribution of charge/electron cloud OR non-linear/bent/V-shaped/angular shape «due to lone pairs» OR polar bonds/dipoles do not cancel out ✓	<i>M2:</i> <i>Do not accept “net/overall dipole moment” without further explanation.</i> <i>Accept “non-symmetrical «shape/distribution of charge»”.</i>	2
7.	a	ii	number of electrons increases ✓ London/dispersion/instantaneous induced dipole-induced dipole forces increase ✓	<i>M1: Accept “M_r/A_r increases” or “molecules become larger in size/mass/surface area”.</i>	2
7.	b		<i>Electron domain geometry:</i> tetrahedral ✓ <i>Molecular geometry:</i> bent/V-shaped/angular ✓	<i>Both marks can be awarded for clear diagrams. Electron domain geometry requires a 3-D diagram showing the tetrahedral arrangement.</i>	2

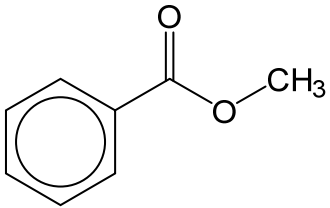
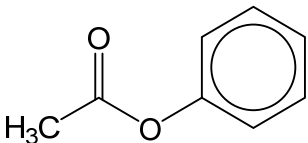
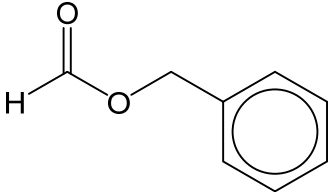
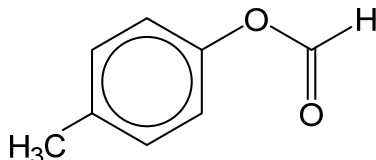
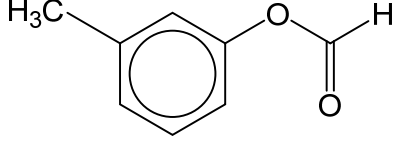
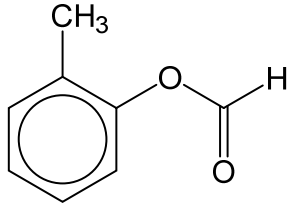
Question			Answers			Notes	Total
7.	c	i	Structure:	I	II	Award [1] for any two correctly filled cells.	2
			O atom labelled (1)	0	«+»1		
			O atom labelled (2)	0	–1		
7.	c	ii	structure I AND no formal charges OR structure I AND no charge transfer «between atoms» ✓				1
7.	d		O ₃ has bond between single and double bond AND O ₂ has double bond OR O ₃ has bond order of 1.5 AND O ₂ has bond order of 2 OR bond in O ₃ is weaker/longer than in O ₂ ✓ O ₃ requires longer wavelength ✓			M1: Do not accept “ozone has one single and one double bond”.	2
7.	e		CO ₂ «non-polar» «weak» London/dispersion forces/instantaneous induced dipole-induced dipole forces between molecules ✓ SiO ₂ network/lattice/3D/giant «covalent» structure ✓			M1: The concept of “between” is essential.	2

Question			Answers	Notes	Total
8.	a		<i>Physical evidence:</i> equal C–C bond «lengths/strengths» OR regular hexagon OR «all» C–C have bond order of 1.5 OR «all» C–C intermediate between single and double bonds ✓ <i>Chemical evidence:</i> undergoes substitution reaction «more readily than addition» OR does not discolour/react with bromine water OR substitution forms only one isomer for 1,2-disubstitution «presence of alternate double bonds would form two isomers» OR more stable than expected «compared to hypothetical molecule cyclohexa-1,3,5-triene» OR enthalpy change of hydrogenation/combustion is less exothermic than predicted «for cyclohexa-1,3,5-triene» ✓	<i>M1:</i> Accept “all C–C–C bond angles are equal”.	2

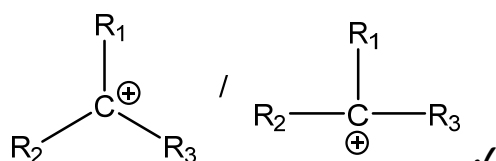
Question			Answers	Notes	Total
8.	b	i	$3\text{CH}_3\text{CH}_2\text{CH}_2\text{OH (l)} + \text{Cr}_2\text{O}_7^{2-} \text{ (aq)} + 8\text{H}^+ \text{ (aq)} \rightarrow 3\text{CH}_3\text{CH}_2\text{CHO (aq)} + 2\text{Cr}^{3+} \text{ (aq)} + 7\text{H}_2\text{O (l)}$ correct reactants and products ✓ balanced equation ✓		2
8	b	ii	<i>Aldehyde:</i> by distillation «removed from reaction mixture as soon as formed» ✓ <i>Carboxylic acid:</i> «heat mixture under» reflux «to achieve complete oxidation to –COOH» ✓	Accept clear diagrams or descriptions of the processes.	2
8.	c	i	« $\frac{136}{48 + 4 + 16} = 2$ » $\text{C}_8\text{H}_8\text{O}_2$ ✓		1
8.	c	ii	A: C–H «in alkanes, alkenes, arenes» AND B: C=O «in aldehydes, ketones, carboxylic acids and esters» ✓		1

(continued...)

(Question 8c continued)

Question			Answers	Notes	Total
8.	c	iii	Any two of:  OR $\text{C}_6\text{H}_5\text{COOCH}_3$ ✓  OR $\text{CH}_3\text{COOC}_6\text{H}_5$ ✓  OR $\text{HCOOCH}_2\text{C}_6\text{H}_5$ ✓	<i>Do not penalize use of Kekule structures for the phenyl group.</i> <i>Accept the following structures:</i>    <i>Award [1 max] for two correct aliphatic/linear esters with the molecular formula $\text{C}_8\text{H}_8\text{O}_2$.</i>	2
8.	c	iv	$\text{C}_6\text{H}_5\text{COOCH}_3$ «signal at 4 ppm (3.7–4.8 range in data table) due to alkyl group on ester» ✓		1

Question			Answers	Notes	Total
9.	a	i	$\begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{H} & & \text{O} & & \text{H} \\ & & & & & & & & & \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{H} \\ & & & & & & & & & \\ & \text{H} & & \text{H} & & \text{H} & & & & \text{H} \end{array}$ / $\begin{array}{c} \text{O} \\ \\ \text{H}_3\text{C} - \text{CH}_2 - \text{CH}_2 - \text{C} - \text{CH}_3 \end{array}$ ✓ $\begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{O} & & \text{H} & & \text{H} \\ & & & & & & & & & \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{H} \\ & & & & & & & & & \\ & \text{H} & & \text{H} & & & & \text{H} & & \text{H} \end{array}$ / $\begin{array}{c} \text{O} \\ \\ \text{CH}_3 - \text{CH}_2 - \text{C} - \text{CH}_2 - \text{CH}_3 \end{array}$ ✓	Accept condensed formulas.	2
9.	a	ii	A: $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$ AND «peak at» 29 due to $(\text{CH}_3\text{CH}_2)^+ / (\text{C}_2\text{H}_5)^+ / (\text{M} - \text{CH}_3\text{CH}_2\text{CO})^+$ OR $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$ AND «peak at» 57 due to $(\text{CH}_3\text{CH}_2\text{CO})^+ / (\text{M} - \text{CH}_3\text{CH}_2)^+ / (\text{M} - \text{C}_2\text{H}_5)^+$ ✓ B: $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$ AND «peak at» 43 due to $(\text{CH}_3\text{CH}_2\text{CH}_2)^+ / (\text{CH}_3\text{CO})^+ / (\text{C}_2\text{H}_3\text{O})^+ / (\text{M} - \text{CH}_3\text{CO})^+$ ✓	Penalize missing “+” sign once only. Accept “$\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$ by elimination since fragment CH_3CO is not listed” for M2.	2

Question			Answers	Notes	Total
9.	b	i	heterolytic/heterolysis ✓		1
9.	b	ii	polar protic ✓		1
9.	b	iii	 Shape: triangular/trigonal planar ✓		2
9.	b	iv	«around» 50 % «each» OR similar/equal percentages ✓ nucleophile can attack from either side «of the planar carbocation» ✓	Accept “racemic mixture/racemate” for M1.	2
9.	c		Stage one: $\text{C}_6\text{H}_5\text{NO}_2(\text{l}) + 3\text{Sn}(\text{s}) + 7\text{H}^+(\text{aq}) \rightarrow \text{C}_6\text{H}_5\text{NH}_3^+(\text{aq}) + 3\text{Sn}^{2+}(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$ ✓ Stage two: $\text{C}_6\text{H}_5\text{NH}_3^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{C}_6\text{H}_5\text{NH}_2(\text{l}) + \text{H}_2\text{O}(\text{l})$ ✓		2